Pharma Mind: AI for smarter Drug Discovery

Submitted in partial fulfilment of the requirements of the degree of

BACHELOR OF ENGINEERING

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CERTIFICATE

This is to certify that the project entitled **"Pharma Mind: AI for smarter Drug Discovery"** is a bonafide work of **Aditya Rawat (22106022), Varun Raut (22106106), Pratik Redekar (22106029), Sarang Sawant (22106080)** submitted to the University of Mumbai in partial fulfilment of the requirement for the award of the degree of **Bachelor of Engineering** in **Computer Science & Engineering (Artificial Intelligence & Machine Learning).**

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Project Report Approval for B.E.

This project report entitled ***"Pharma Mind: AI for smarter Drug Discovery"*** by ***"Aditya Rawat, Varun Raut, Pratik Redekar, Sarang Sawant"*** is approved for the degree of ***Bachelor of Engineering*** in ***Computer Science & Engineering (Artificial Intelligence & Machine Learning), 2025-26***.

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Place:

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We declare that this written submission represents my ideas in my own words and where others' ideas or words have been included, I have adequately cited and referenced the original sources. I also declare that I have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in my submission. I understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

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Abstract

Drug discovery is a complex, time-intensive, and costly process that requires accurate prediction of compound activity against protein targets. Recent advances in artificial intelligence (AI) and graph-based machine learning have enabled the representation of chemical molecules as structured graphs, allowing models to directly leverage molecular topology and atomic features. In this work, we present an end-to-end machine learning pipeline that integrates molecular graphs derived from SMILES strings with protein target embeddings to predict bioactivity values. The system employs Graph Convolutional Networks (GCNs) for learning molecular representations, combined with target-conditioned embeddings to capture compound–target interactions. Two variants of the pipeline are explored: a classification approach for predicting binary activity (active/inactive) and a regression approach for estimating continuous activity values.

Keywords:

Drug Discovery, Artificial Intelligence (AI), Graph Convolutional Network (GCN),Molecular Graphs, SMILES Representation, Protein Targets, Machine Learning in Healthcare.

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ABBREVATION

ADMET	*Absorption, Distribution, Metabolism, Excretion, and Toxicity*
QSAR	*Quantitative Structure–Activity Relationship*
SMILES	*Simplified Molecular Input Line Entry System*
DTI	*Drug–Target Interaction*
PubChem	*Public Chemical Database*
ChEMBL	*Chemical Molecule Bioactivity Database*
DrugBank	*Drug Information Database*
BindingDB	*Binding Database for Protein–Ligand Interactions*

Chapter 1

INTRODUCTION

Drug discovery is a critical process in pharmaceutical research, aimed at identifying novel compounds that can interact effectively with biological targets such as proteins, enzymes, or receptors. Traditionally, this process relies heavily on experimental screening, where thousands of candidate molecules are tested in wet-lab settings to evaluate their activity and safety profiles. While effective, this approach is both time-intensive and cost-prohibitive, often requiring several years and billions of dollars before a potential drug reaches the clinical trial stage.

In recent years, Artificial Intelligence (AI) and Machine Learning (ML) have emerged as transformative technologies in the life sciences, offering computational methods that can accelerate and optimize various stages of drug discovery. By leveraging large-scale chemical and biological datasets, AI-based models can learn patterns of compound–target interactions and predict the likelihood of bioactivity without the need for exhaustive laboratory testing. This reduces not only the time and cost but also enhances the efficiency of virtual screening pipelines.

One promising line of research is the application of Graph Neural Networks (GNNs), particularly Graph Convolutional Networks (GCNs), in modelling molecular structures. Unlike traditional descriptor-based models, which depend on manually engineered chemical features, GCNs operate directly on the graph representation of molecules.

Chapter 2

LITERATURE SURVEY

[1] ***Tuan Anh Nguyen, Thien Huynh, and Hiroshi Mamitsuka.*** **"Artificial Intelligence and Machine Learning in Drug Discovery and Development" (2022)**

This paper explores how Artificial Intelligence (AI) and Machine Learning (ML) are transforming the drug discovery and development pipeline. It highlights the use of advanced algorithms such as deep learning, reinforcement learning, and natural language processing for target identification, molecular screening, and clinical data analysis.

[2] ***Alex Zhavoronkov, Alán Aspuru-Guzik, and Sean Ekins.*** **"Artificial Intelligence in Drug Discovery: Applications and Techniques" (2021)**

This paper presents a comprehensive overview of the applications and technical approaches of Artificial Intelligence (AI) in the field of drug discovery. It discusses how machine learning (ML) and deep learning (DL) models—such as convolutional neural networks (CNNs), recurrent neural networks (RNNs), and generative models—are utilized for molecular property prediction, virtual screening, and de novo drug design.

[3] ***H. Shin, S. Park, and J. Lee.*** **"DeepGS: Deep Representation Learning of Graphs and Sequences for Drug–Target Binding Affinity Prediction" (2020)**

This paper introduces DeepGS, a hybrid deep learning framework that integrates Graph Convolutional Networks (GCNs) for molecular graph encoding and Convolutional Neural Networks (CNNs) for protein sequence modeling. By combining these representations, the model predicts drug–target binding affinities with high precision on benchmark datasets such as KIBA and Davis.

[4] ***Liang Chen, Priya Natarajan, and Markus Hoffmann.*** **"DeepLead Optimization: Leveraging Generative AI for Structural Modification" (2024)**

This paper introduces DeepLead Optimization, a Generative AI–based framework designed for lead compound optimization in drug discovery. The model leverages Variational Autoencoders (VAEs), Generative Adversarial Networks (GANs), and Reinforcement Learning (RL) to perform structural modifications that enhance drug-likeness, bioactivity, and ADMET properties.

[5] ***Rui Zhang, Ming Li, and Daniela Cruz.*** **"Knowledge-augmented Graph Machine Learning for Drug Discovery: A Survey" (2023)**

This survey paper reviews advancements in knowledge-augmented graph machine learning (Graph ML) techniques applied to drug discovery. It explores how Graph Neural Networks (GNNs)—such as Graph Convolutional Networks (GCNs) and Graph Attention Networks (GATs)—can be integrated with biomedical knowledge graphs to model complex drug–disease–target relationships. The authors analyze various datasets like DrugBank, ChEMBL, and Hetionet, emphasizing how combining domain knowledge with graph-based learning improves interpretability and predictive power.

[6] Alexandre Blanco-González, Alfonso Cabezón, Alejandro Seco-González, Daniel Conde-Torres, Paula Antelo-Riveiro, Ángel Piñeiro, and Rebeca Garcia-Fandino. ***"The Role of AI in Drug Discovery: Challenges, Opportunities, and Strategies"*** **(2021)**

This paper provides a comprehensive overview of Artificial Intelligence (AI) applications throughout the drug discovery pipeline, from target identification to lead optimization and clinical trials. It emphasizes how AI techniques such as Machine Learning (ML), Deep Learning (DL), and Natural Language Processing (NLP)accelerate the discovery process by identifying novel molecular structures, predicting bioactivity, and optimizing pharmacokinetics. The study discusses several datasets including PubChem, ChEMBL, and DrugBank used for training and validation of AI models.

[7] ***Anselm Bender. "Artificial Intelligence in Drug Discovery: What Is Realistic, What Is Illusory?"*** **(2021)**

This article critically examines the current and potential applications of Artificial Intelligence (AI) in drug discovery, highlighting both its achievements and limitations. It discusses how AI has been successfully applied in areas such as image recognition and natural language processing but points out that comparable advances in drug discovery are rare. The author emphasizes the need for realistic expectations and careful consideration of the challenges and complexities involved in integrating AI into the drug discovery process.

[8] Tristan S. Maurer, Martin Edwards, David Hepworth, Patrick Verhoest, and Charlotte M.N. Allerton. ***"Designing Small Molecules for Therapeutic Success: A Contemporary Perspective"*** **(2021)**

This paper presents a contemporary overview of strategies for designing small molecules with therapeutic potential. It explores computational approaches such as structure-based drug design, ligand-based design, and AI-driven modeling to optimize pharmacological properties and drug-likeness. The authors emphasize the importance of integrating chemical, biological, and clinical knowledge for effective molecular design.

[9] Rocío Mercado, Tobias Rastemo, Edvard Lindelöf, Günter Klambauer, Ola Engkvist, Hongming Chen, and Esben Jannik Bjerrum. ***"Practical Notes on Building Molecular Graph Generative Models"*** **(2021)**

This paper offers practical guidance for building molecular graph generative models, a growing area in computational drug discovery. It reviews methodologies such as VAEs, GANs, and GNNs, highlighting their strengths and weaknesses for de novo molecular generation. The authors address key implementation challenges including dataset preparation, model evaluation, and chemical validity enforcement.

[10] M. Besharatifard and F. Vafaee. *"A Review on Graph Neural Networks for Predicting Synergistic Drug Combinations"* (2024)

This review examines the application of Graph Neural Networks (GNNs) in predicting synergistic drug combinations. It discusses the advantages of GNNs in representing complex molecular interactions, the strengths and weaknesses of various GNN models, and the datasets used for training.

Year	Title	Domain	Algorithm	Dataset	Gap
2022	Artificial Intelligence and Machine Learning in Drug Discovery and Development	Drug discovery & development using Artificial Intelligence (AI) and Machine Learning (ML)	- Deep Learning (DL) - Neural Networks - Random Forest -SVM - Reinforcement Learning (RL) -Natural Language Processing (NLP)	-Public datasets (e.g., PubChem, ChEMBL, Protein DrugBank)	-Limited availability and quality of labeled biomedical data. - High cost and Complexity. - Lack of Interpretability in AI/ML models.
2021	Artificial Intelligence in Drug Discovery: Applications and Techniques	AI applications in Drug Discovery & Development	- Machine Learning - Deep Learning (DL) - Reinforcement Learning (RL) - Natural Language Processing (NLP)	-Public repositorielike ChEMBL, DrugBankPubChem, Protein Data Bank	- Data scarcity and imbalance for rare diseases -Model issues. -Transferability/ generalization across different drug classes is limited.
2020	DeepGS: Deep Representation Learning of Graphs and Sequences for Drug–Target Binding Affinity Prediction	ComputationDrug Discovery – Drug–Target Interaction (DTI) Prediction	- Proposed DeepGS model combining: • Graph Convolutional Networks (GCN) for drug molecular graphs • Convolutional Neural Networks (CNN) for protein sequences - Joint representation	- KIBA dataset (drug–target binding affinity data)	- Model relies heavily on benchmark datasets, which may not fully reflect real-world complexity.

			learning for binding affinity prediction		
2024	DeepLead Optimization: Leveraging Generative AI for Structural Modification	AI-driven Drug Design – Lead Optimization & Molecular Generation	- Multi-objective optimization (drug-likeness, ADMET properties) - Structure-based molecular optimization	- ADMET property datasets for validation	- Regulatory and safety assessments of AI-designed drugs are underdeveloped.
2023	Knowledge-augmented Graph Machine Learning for Drug Discovery: A Survey	AI for Drug Discovery – Graph Machine Learning (Graph ML) with Knowledge Integration	- Graph Neural Networks (GNNs): Graph Convolutional Networks (GCNs), Graph Attention Networks (GATs)	- Drug–target interaction datasets (DrugBank, ChEMBL, BindingDB, PubChem)	- Limited benchmarks for knowledge-augmented GNNs in drug discovery. - Interpretability of Graph ML models is still weak.
2023	The Role of AI in Drug Discovery: Challenges, Opportunities, and Strategies	AI-driven Drug Discovery and Development	ML, DL, NLP-based predictive modeling	PubChem, ChEMBL, DrugBank	Despite notable progress the integration of AI model with experimental validation remains limited
2020	"Artificial Intelligence in Drug Discovery: What Is Realistic, What Are Illusions	Critical analysis of AI Application in Drug Discovery.	Review and analysis of existing AI techniques	ChEMBL, PubChem, DrugBank	Limited integration of AI in clinically relevant decision-making
2021	Designing Small Molecules for Therapeutic Success.	small-molecule drug design for	molecular docking, QSAR models, and AI-assisted	proprietary pharmaceutical compound datasets.	Limited integration of AI models with medicinal

		therapeutic applications.	molecular optimization.		chemistry workflows.
2021	Practical Notes on Building Molecular Graph Generative Models	Generative modeling of molecular graphs for drug discovery	graph-based generative models Variational Autoencoders (VAEs) Generative Adversarial Networks (GANs)	ZINC, ChEMBL proprietary chemical compound	Challenges in ensuring chemical validity and synthetic feasibility of generated molecules
2024	*"A Review on Graph Neural Networks for Predicting Synergistic Drug Combinations"*	AI and Graph Neural Networks for predicting synergistic drug combinations.	Graph Neural Networks (GNNs), deep learning, and network-based learning approaches.	Drug CombDB, Synergy-Finder, and biomedical molecular interaction datasets.	Limited standardization of GNN models for drug combination prediction

Table 2.1: Use-case of Research Papers

Chapter 3

LIMITATIONS OF EXISTING SYSTEMS

Classical ML (QSAR/fingerprints):

- Handcrafted features → limited representation of molecular complexity.
- Struggles with unseen scaffolds and novel chemistries.

SMILES-based DL models (RNNs/Transformers):

- Miss full graph topology, stereochemistry & 3D conformation.
- Sensitive to SMILES syntax variations.

Graph Neural Networks (GCN/MPNN):

- Strong at structure learning, but data hungry.
- Lack interpretability → "black box" decisions not trusted in pharma.

Observations:

Problem not yet completely solved:

- No existing model addresses *all* challenges → data quality, scalability, interpretability, and generalization still open issues.

Technologies with promise:

- GNNs/MPNNs show best trade-off between accuracy and scalability.
- Pretraining learning methods improve data efficiency.

Improvement possibilities:

- Better data preprocessing and balanced datasets.
- Hybrid models combining graph learning + chemical knowledge.
- Explainable AI tools for trust and regulatory compliance.

Chapter 4

PROBLEM STATEMENT, OBJECTIVES, SCOPE

4.1 Problem Statement

Drug discovery is a complex and data-intensive process that involves identifying potential compounds capable of interacting effectively with biological targets such as proteins or enzymes. Traditional experimental methods are time-consuming, costly, and limited by the scope of laboratory testing.

To design and implement an AI-driven Graph Neural Network (GCN)–based system that predicts molecular bioactivity and properties from graph representations of compounds, thereby improving the efficiency, scalability, and interpretability of the drug discovery process.

This approach aims to overcome limitations of traditional QSAR and SMILES-based models by leveraging graph-based learning for a deeper understanding of molecular topology and drug–target interactions.

4.2 Objectives

1. To accelerate drug discovery using Artificial Intelligence and Machine Learning techniques for predictive modeling of molecular interactions.
2. To implement Graph Neural Networks (GNNs) for representing and learning from molecular graph structures directly rather than relying on handcrafted chemical descriptors.
3. To enhance prediction accuracy for compound activity and bio-property estimation using integrated graph and sequence learning.
4. To reduce cost and experimentation time by enabling in-silico screening before wet-lab validation.
5. To improve interpretability and scalability of AI-based drug discovery models through knowledge-augmented graph learning and explainable AI techniques.
6. To utilize publicly available datasets (such as ChEMBL, DrugBank, PubChem, and BindingDB)

4.3 Scope

- **PredictiveModeling:** Development of machine learning and graph-based models capable of predicting molecular bioactivity, toxicity, and pharmacological properties.
- **Graph-Based Representation:** Use of Graph Convolutional Networks (GCNs) to learn molecular representations from atomic structures and bonding information.
- **Target-ConditionedLearning:** Integration of protein target embeddings for compound–target interaction prediction.
- **DatasetUtilization:** Leveraging publicly available molecular and biological databases such as **DrugBank**, **ChEMBL**, **PubChem**, and **BindingDB** for training and validation.

Chapter 5

PROPOSED SYSTEM

Data Flow Diagram:

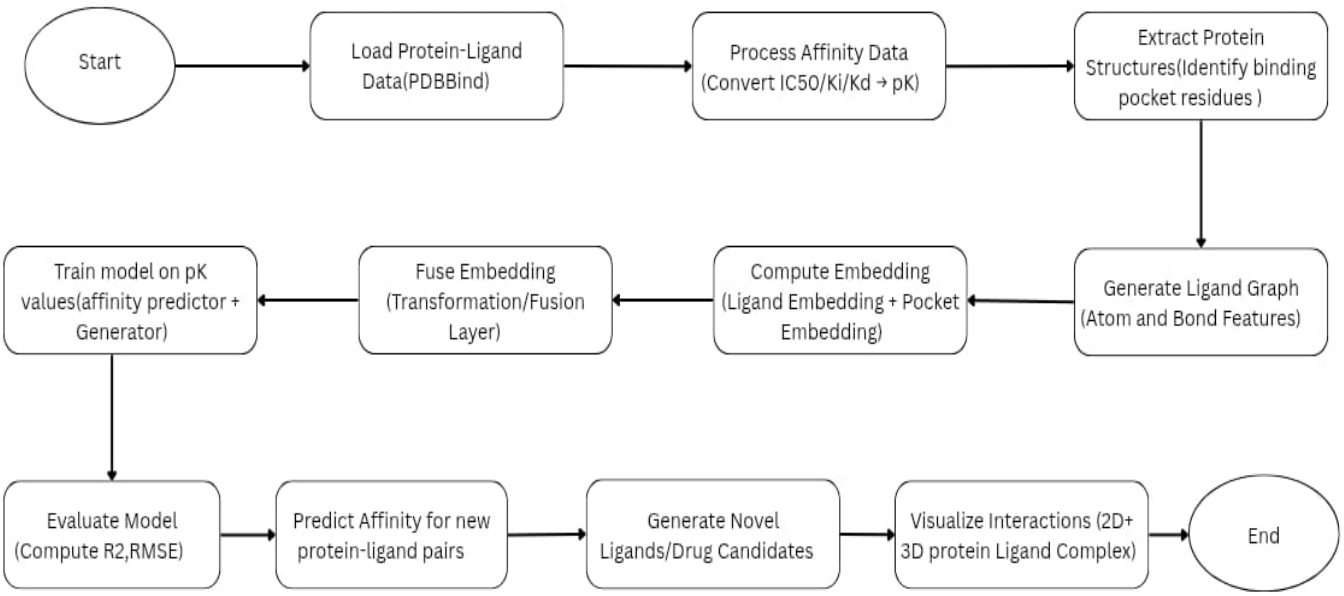


Figure 5.1 Activity Diagram

5.1 System Overview

The proposed system, Pharma Mind, is an AI-driven framework designed to accelerate the drug discovery process using Graph Neural Networks (GNNs) and embedding-based models. The system predicts binding affinity between proteins and ligands by learning from molecular graphs and protein structural data. It also assists in generating potential drug candidates and visualizing molecular interactions in both 2D and 3D formats.

System Workflow Steps

1. **LoadProtein–LigandData**:
   Retrieve datasets such as PDBBind, containing protein–ligand complexes and affinity data.

2. **ProcessAffinityData:**
   Convert experimental values (IC50, Ki, Kd) into standardized pK scores for training consistency.

3. **ExtractProteinStructures:**
   Identify binding pocket residues within protein structures that interact with ligands.

4. **GenerateLigandGraph:**
   Represent each molecule as a graph with atoms as nodes and bonds as edges; extract atomic and bond-level features.

5. **ComputeEmbeddings:**
   Generate ligand embeddings and pocket embeddings to capture chemical and structural characteristics.

6. **FuseEmbeddings**:
   Combine embeddings through a fusion layer to model ligand–protein interactions effectively.

7. **TrainModel:**
   Train a Graph Convolutional Network (GCN) on pK values to learn patterns of molecular affinity.

8. **EvaluateModel:**
   Assess performance using metrics such as R^2 and RMSE to validate prediction accuracy.

9. **PredictNewAffinities:**
   Apply the trained model to predict affinities for unseen protein–ligand pairs.

10. **GenerateNovelCompounds:**
    Use the model's generator to suggest new ligands or drug candidates with improved bioactivity.

11. **VisualizeInteractions:**
    Display 2D and 3D protein–ligand complexes to analyse binding interactions and validate molecular behaviour.

Chapter 6

EXPERIMENTAL SETUP

Software Requirements: -

- **Python 3.10+:** Core programming language used for model development.
- **Jupyter Notebook / Google Colab:** Environment for running and testing AI models interactively.
- **PyTorch / TensorFlow:** Frameworks used to implement Graph Neural Networks (GCNs).

Hardware Requirements:

1. **Processor:** Minimum of Intel Core i5 or equivalent (Quad-core) for local development and processing.
2. **RAM:** At least 16 GB for smooth running of applications, especially when handling large datasets for report generation.
3. **Storage:** Minimum of 256 GB SSD for fast read/write operations and storage of intermediate data, models, and processed files.

Operating Environment:

Platform: Windows 10 and Google Colab (cloud environment)

Development Environment:

Python 3.10+: Required for Jupyter Notebook, PyTorch, and machine learning tasks.

Machine Learning & NLP Libraries:

PyTorch: For implementing and training Graph Convolutional Networks.

RDKit: For molecular graph generation and feature extraction.

Scikit-learn: For preprocessing and evaluation (accuracy, precision, recall, RMSE).

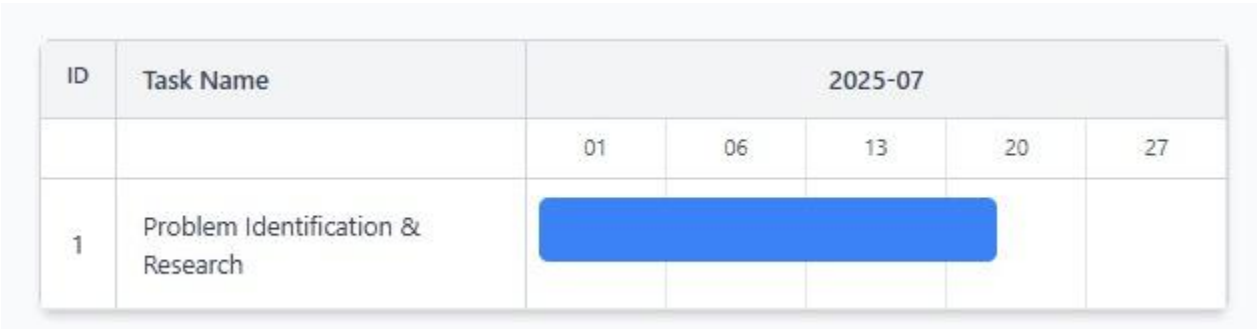
Pandas / NumPy: For dataset handling and vectorized computations.

Matplotlib / Seaborn: For plotting learning curves and molecular visualizations.

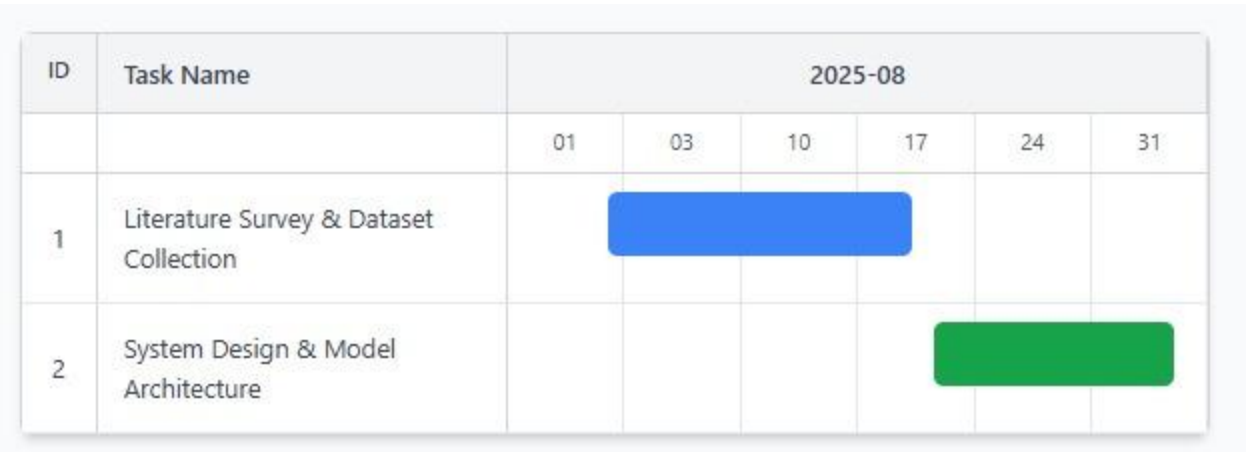
Chapter 7

PROJECT PLANNING

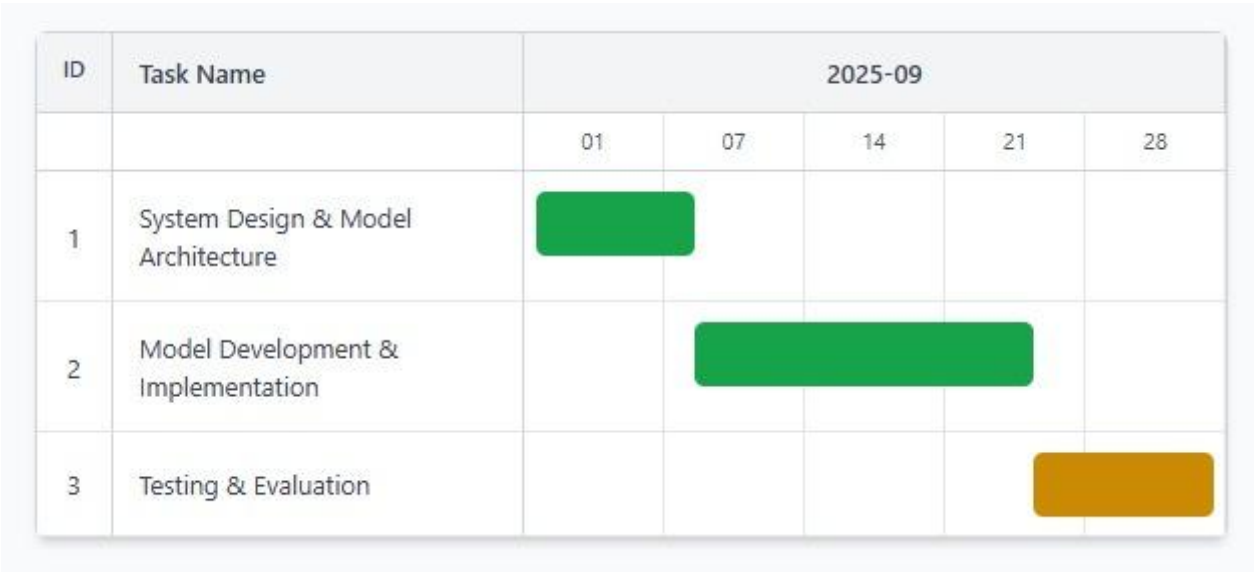
Gantt Chart:



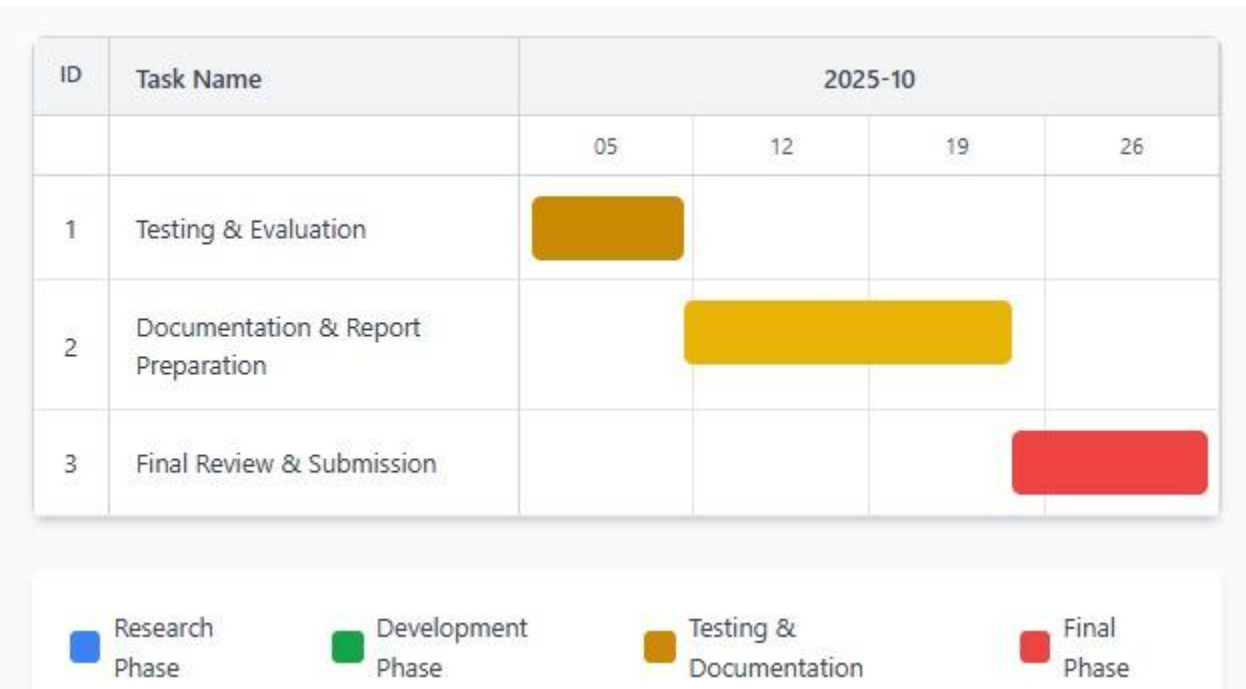
Graphs: 7.1 July (Gantt Chart)



Graphs: 7.2 August (Gantt Chart)



Graphs: 7.3 September (Gantt Chart)



Graphs: 7.4 October Gantt (Chart)

Chapter 8

EXPECTED OUTCOME

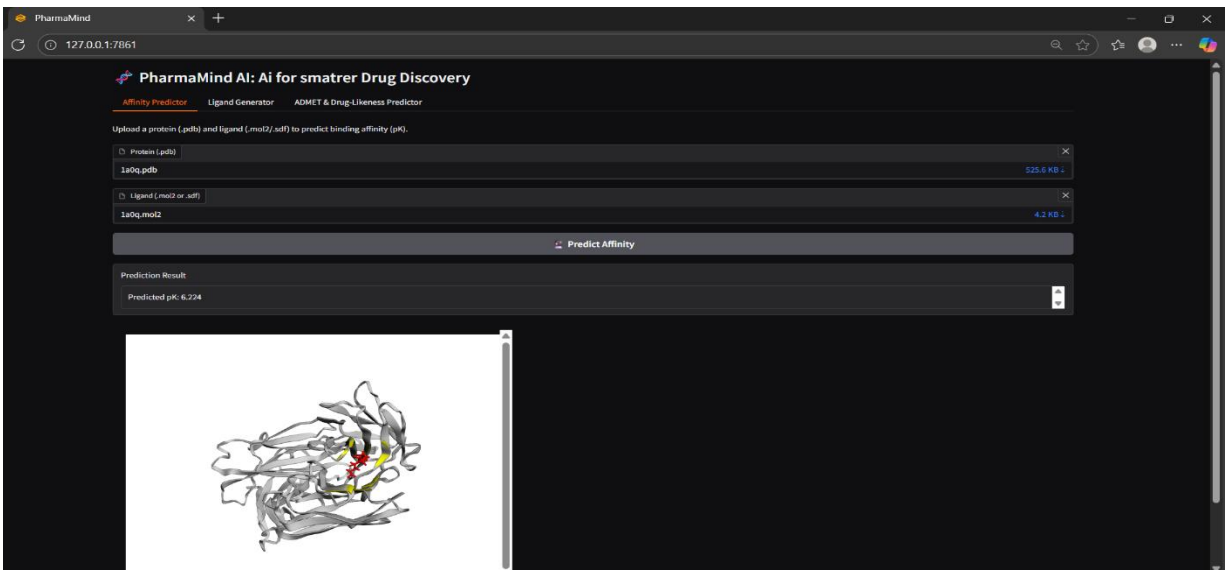


Figure 8.1: Predicting Binding Affinity

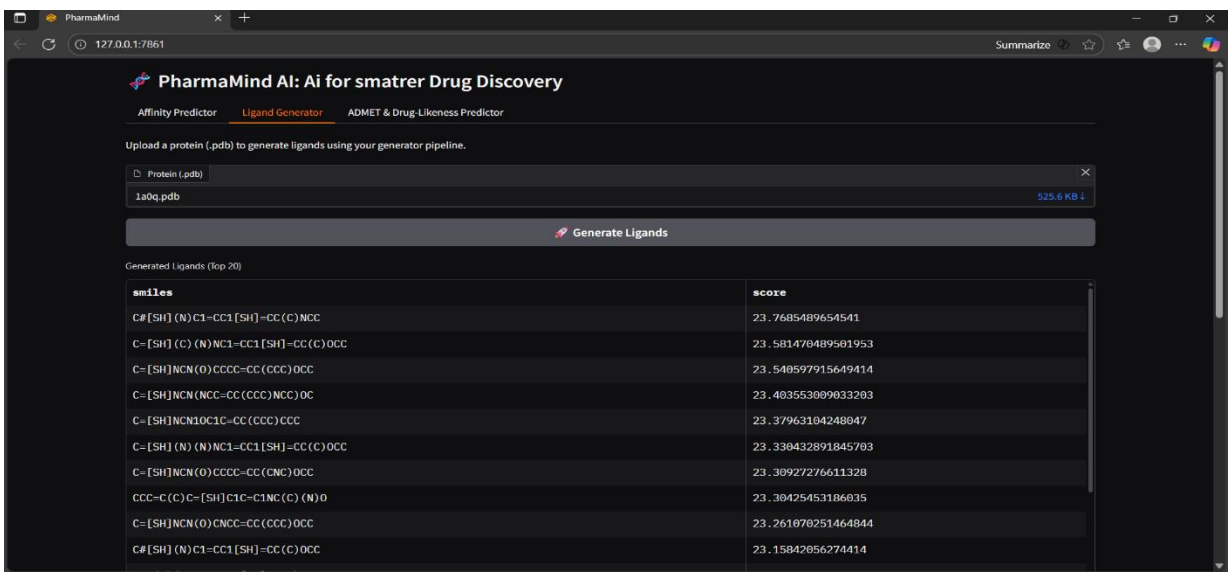


Figure 8.2: Generating Ligands

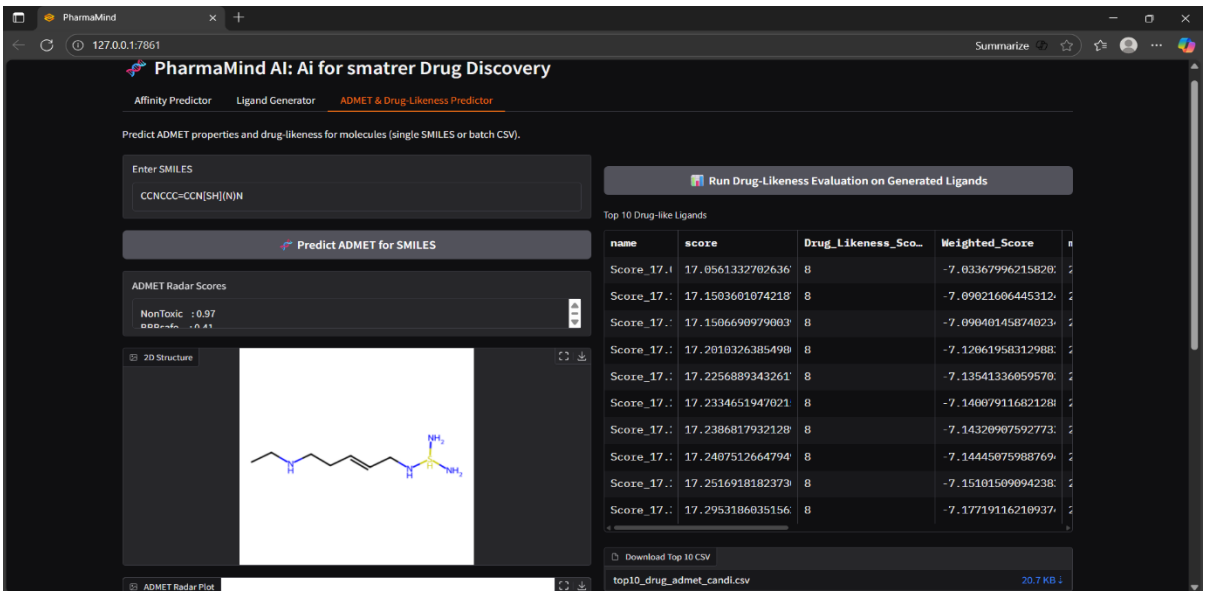


Figure 8.3:Admet Prediction

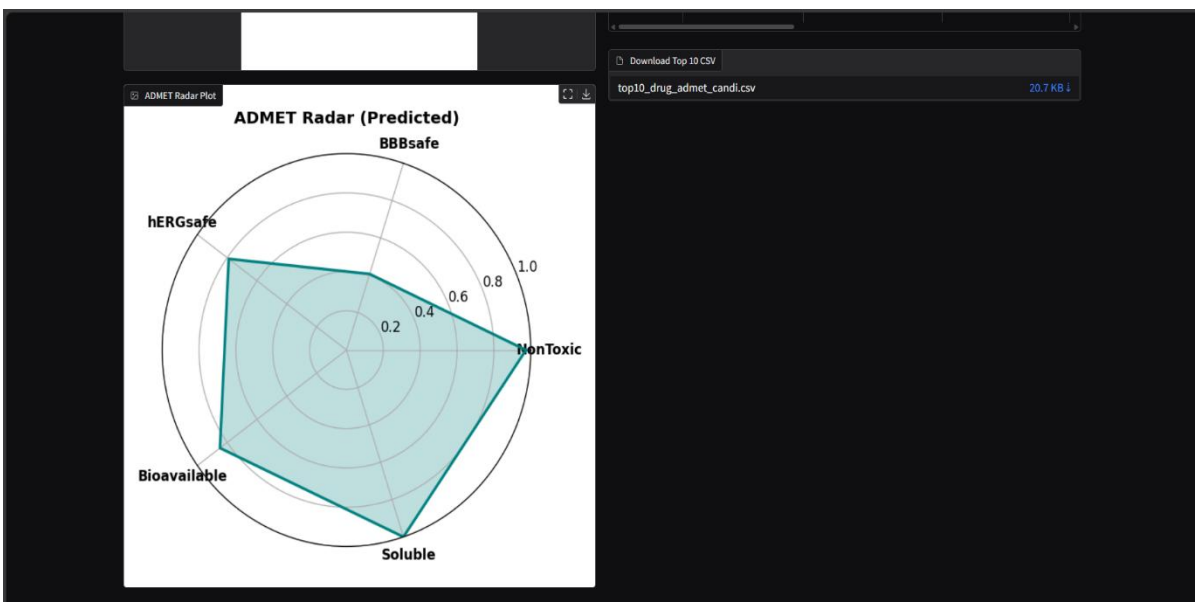


Figure 8.4:Admet Radar Plot

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